Data Science Programming

D7054E Group project

Project Title: Predicting House Prices Using Machine Learning

Submitted by: Group 38

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1. Business/Real-life Problem

Accurate house price prediction is crucial for buyers, sellers, and real estate professionals to make informed decisions. Traditional valuation methods may not capture complex relationships between property features and market prices. This project aims to build a predictive model that estimates house prices based on various property attributes.

2. Aim/Hypothesis

To develop a machine learning model that can predict house prices based on features such as location, size, number of rooms, and other property characteristics.
Hypothesis: Property features like square footage, location, and number of rooms significantly influence house prices and can be used to build an accurate predictive model.

3. Dataset Details

- **Source: Kaggle –** House Prices: Advanced Regression Techniques
- **Link:** House Prices - Advanced Regression Techniques | Kaggle
- **Description:** The dataset includes 79 explanatory variables describing (mostly) residential homes in Ames, Iowa. It contains 1,460 observations with features such as lot area, year built, number of bedrooms, and sale price.

4. Algorithms/Techniques Considered

- Data preprocessing: Handling missing values, encoding categorical variables, feature scaling
- Exploratory Data Analysis (EDA): Correlation analysis, distribution plots, outlier detection
- Feature engineering: Creating new features, dimensionality reduction (e.g., PCA)
- Model training: Linear Regression, Ridge/Lasso Regression, Random Forest, XGBoost
- Model evaluation: RMSE, R^2 score, cross-validation
- Visualization: Feature importance plots, residual plots, predicted vs. actual price graphs
- Ethical considerations: Bias in pricing models, fairness in real estate valuation